Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Image processing**	
Experiment:-7	**Date:15-08-2026**	**Enrollment No:92400133063**

Aim:-image processing

Code:-

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Read image directly in grayscale
src_image = cv2.imread('1.png', cv2.IMREAD_GRAYSCALE)

# Check if image is loaded
if src_image is None:
    print("Image not found!")
    exit()

# ------------------------
# 1. Image Negation
# ------------------------
negative_image = 255 - src_image

# ------------------------
# 2. Thresholding
# ------------------------
_, thresholded_image = cv2.threshold(src_image, 128, 255,
cv2.THRESH_BINARY)

# ------------------------
# 3. Gamma Correction
# ------------------------
gamma = 2.0

# Normalize image to range [0,1]
normalized_image = src_image / 255.0
```

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```
# Apply gamma correction
gamma_corrected_image = np.power(normalized_image, 1/gamma)

# Convert back to range [0,255]
gamma_corrected_image = np.uint8(gamma_corrected_image * 255)

# ------------------------
# Display Images
# ------------------------
plt.figure(figsize=(15,5))

plt.subplot(1,4,1)
plt.imshow(src_image, cmap='gray')
plt.title("Original")
plt.axis('off')

plt.subplot(1,4,2)
plt.imshow(negative_image, cmap='gray')
plt.title("Negative")
plt.axis('off')

plt.subplot(1,4,3)
plt.imshow(thresholded_image, cmap='gray')
plt.title("Threshold")
plt.axis('off')

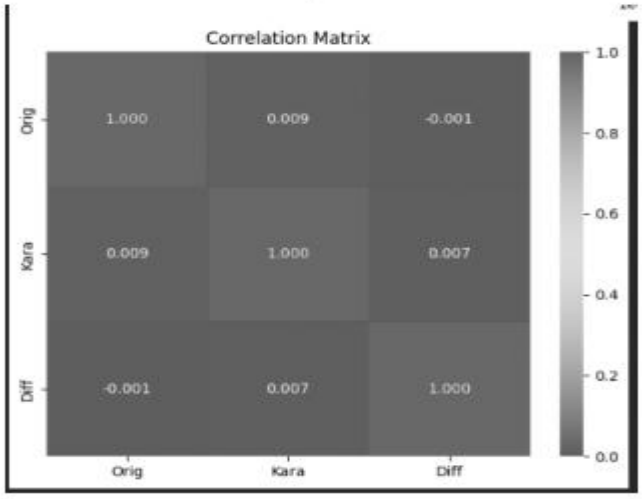
plt.subplot(1,4,4)
plt.imshow(gamma_corrected_image, cmap='gray')
plt.title("Gamma")
plt.axis('off')

plt.show()
```

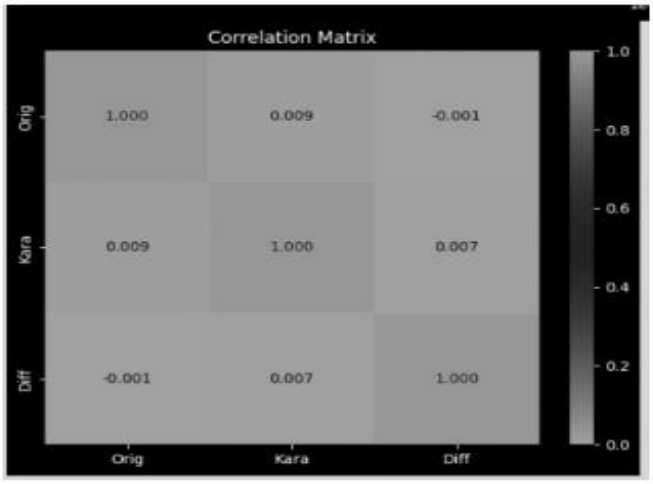
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output:-

Original

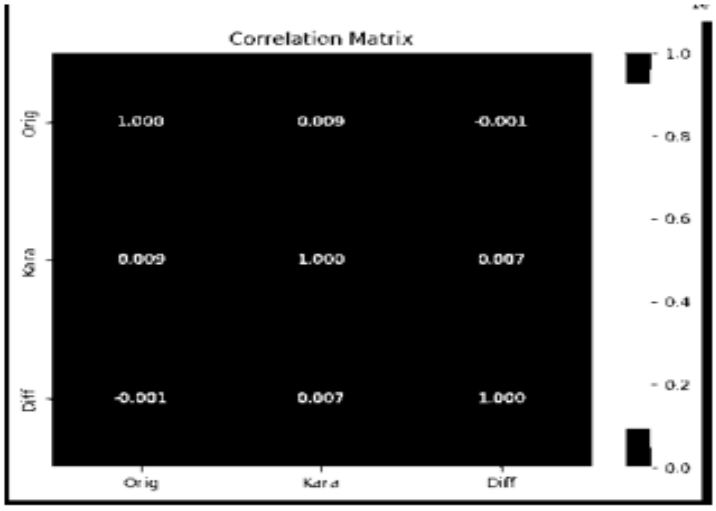


Negative

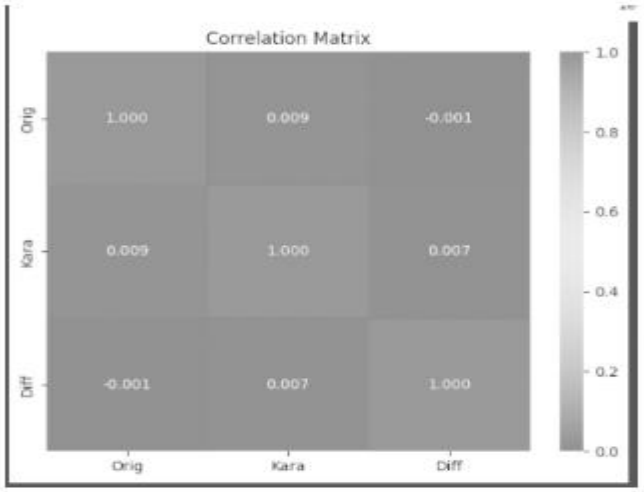


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Threshold



Gamma



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Conclusion:-

Image negation (255 - pixel value) which inverts brightness, binary thresholding (at 128) which converts the image to black and white based on pixel intensity, and gamma correction (γ=2.0) which enhances mid-tone details by adjusting the intensity curve non-linearly. The original grayscale image is loaded and all transformations are applied directly to the pixel array, with results displayed side-by-side for easy comparison. Each technique serves different purposes: negation for photometric inversion, thresholding for segmentation, and gamma correction for brightness/contrast optimization. The code effectively showcases how simple mathematical operations on pixel values can drastically alter image appearance and extract useful information.